

Sagar Sinha

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EDUCATION

MS Computer Science

Dec 2025

Arizona State University

GPA: 3.93/4.0

- **Relevant Coursework:** Data Mining, Applied Regression Analysis, NLP, Data Visualization, Distributed Database Systems

BS Computer Engineering

May 2022

Pandit Deendayal Energy University

GPA: 9.02/10.0

- **Relevant Coursework:** Data Structures, Algorithms, RDBMS, Data Warehousing, Statistics, Machine Learning

PROFESSIONAL EXPERIENCE

Backend Developer (AWS) Intern, Mindsparkd

Mar 2026 – Present

- Built a **System Health data workflow** with **AWS Lambda, DynamoDB, Aurora PostgreSQL, API Gateway**, and **CloudWatch** to generate rolling-window anomaly insights from Kairos interaction logs.
- Improved sidebar data retrieval performance by **40%** by optimizing **Aurora PostgreSQL** query paths, removing redundant joins, and adding targeted indexes.
- Analyzed **serverless data access patterns** using **CloudWatch Insights, RDS metrics**, and **Performance Insights** to identify reliability, timeout, retry, and scaling risks.
- **Technical Skills:** Python, SQL, AWS Lambda, Aurora PostgreSQL, DynamoDB, API Gateway, Step Functions, CloudWatch, Performance Insights, Terraform, Git.

AI & Data Processing Externship, Extern

Sep 2025 – Dec 2025

- Implemented an **OCR indexing engine** that converted **200+ page unstructured mortgage documents** into searchable context for an **AI Q&A workflow**.
- Implemented **hybrid retrieval** with optimized chunking and vector search, achieving **2s median query latency** and **MRR = 0.98** for mortgage document search.
- Added **pipeline observability** with logging for data lineage, document routing, and processing metrics, improving traceability across the mortgage workflow.
- **Technical Skills:** Python, OCR, Vector Search, RAG, Data Indexing, Logging, Data Lineage, Pipeline Observability.

Data Analyst, Arizona State University

Jul 2024 – Dec 2025

- Engineered an **AWS ELT pipeline** to ingest **50K+ Canvas records** using Selenium and BeautifulSoup, staging data in **S3** and loading structured datasets into **Redshift**.
- Built **Tableau dashboards** and KPI reports comparing ACHA/NCHA survey results with national benchmarks, tracking **40+ wellness KPIs** across 3 years.
- Standardized survey reporting with intake templates and scripted **QA checks**, reducing ad-hoc cleaning while surfacing **4 high-risk mental health trends**.
- **Technical Skills:** Python, MySQL, Tableau, AWS, EDA, Data Modeling.

Data Science Researcher, Accredian

Apr 2023 – Jul 2023

- Built **Python feature-engineering pipelines** with Pandas, NumPy, and Torch to support learner engagement analysis for **3,000+ platform users**.
- Performed **root-cause analysis** on recurring data issues, improving production support workflows and contributing to a **90% issue-resolution rate**.
- Refactored legacy scripts into **modular Python components**, reducing technical debt and improving reliability of learner engagement datasets.
- **Technical Skills:** Python, Pandas, NumPy, PyTorch, MySQL, Tableau, EDA, Data Modeling.

RELEVANT PROJECTS

Repairable Reasoning in Discrete Diffusion LLMs

- Engineered a unified evaluation harness to compare Discrete Diffusion (Llada-8B) and Autoregressive (Llama-8B) models across GSM8K, AIME, LogiQA, and GPQA datasets.
- Orchestrated reparability trials revealing that extra compute scales Llada accuracy to 85.0% (13.5% gain) with zero over-repair, compared to high over-repair and stagnation in Llama models.
- **Technical Skills:** Python (Torch), Huggingface, LLM, Supervised Finetuning, Prompting, Inference, Evaluation Metrics

ParkFlow - Geospatial Analytics for Ride Demand Monitoring

- Developed **SQL queries** on a **large structured** amusement park database and orchestrated **Python analyses** to map park-wide visitor flows and ride utilization.
- Executed in-depth statistical **EDA** and implemented **Shewhart-style control-chart** monitoring on ride-level attendance to pinpoint underperforming attractions, improving **anomaly-detection** precision by **25%**.
- Leveraged **geospatial analytics (PySAL – Moran's I)** and produced choropleth maps to guide staffing and resource allocation across **15+ attractions**, accelerating location-based decisions by **20%**.
- **Technical Skills:** Python(Pandas, Matplotlib, Statsmodels, Geopandas, Seaborn), MySQL

HousePriceIQ - Feature Selection & Price Prediction Modeling

- Optimized 80+ sparse features using a two-stage selection workflow (VarianceThreshold, ExtraTrees), effectively filtering noise and identifying primary price drivers.
- Tuned a RandomForest Regressor with custom RMSLE and Adjusted R^2 metrics, leveraging one-hot encoding to mitigate overfitting across heterogeneous data splits.
- **Technical Skills:** Python(Scikit-Learn, FastAPI), Data Visualization, Jupyter Notebook